

Sales Performance Analysis

AI-Assisted Power BI Dashboard Build & Insights Report

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Data source: Sample Sales Data (Kaggle) — 2,823 transaction records, 2003–2005
Tools used: Power BI Desktop, Claude Code, pbi-cli (Power BI MCP-style skills)

Executive Summary

This report summarises a sales performance analysis built end-to-end using an AI-assisted Power BI workflow. Total sales across the dataset reached **\$10.03M** from **307 orders** and **92 customers**, at an average order value of **\$32,680**. The analysis surfaces which product lines, countries, and deal sizes drive the majority of revenue, and tracks performance trends across the 2003–2005 period covered by the dataset.

\$10.03M	307	92	\$32.68K
Total Sales	Total Orders	Total Customers	Avg. Order Value

Methodology

The dashboard was built using an AI-assisted, natural-language-driven workflow rather than manual point-and-click report building, intended as a demonstration of applying AI agents to real BI delivery tasks.

- **Data source:** A public sample sales dataset (Kaggle, kyanyoga/sample-sales-data) was used, containing order-level transactions with customer, product, geography, and date attributes.
- **Tooling:** Claude Code (an AI coding agent) was connected to a live Power BI Desktop session using pbi-cli, a toolkit that exposes Power BI's semantic model and PBIR report format as instructions an AI agent can act on directly.
- **Semantic model:** Tables, relationships, and DAX measures (Total Sales, Total Orders, YTD/QTD/MTD, YoY%, Deal Size breakdown, etc.) were generated from the raw CSV via natural-language prompts, then validated against the source data end-to-end.
- **Report build:** Four report pages (Overview, Product & Deal Size, Geography, Trends & Time Intelligence) were generated as Power BI Report JSON (PBIR), including chart selection, layout, and a custom navy/steel-blue corporate theme.
- **Validation & QA:** Broken visual bindings and a report-load error introduced during theming were identified and corrected using the tool's built-in schema validation, with fixes verified before final review in Power BI Desktop.

One data limitation is noted transparently: the source dataset has no cost/COGS column, so true profit margin could not be calculated. Where list-price comparisons were used as a proxy, this is labelled explicitly rather than presented as genuine margin.

Key Findings

1. Product Line Performance

Classic Cars is the dominant product line, generating **\$3.92M** (39% of total sales) from 33,992 units sold. Vintage Cars (\$1.90M) and Motorcycles (\$1.17M) follow as the next strongest categories. Trains is the smallest contributor at \$226K, suggesting either limited demand or a narrower catalogue within that line.

Product Line	Total Sales	Units Sold	Avg. Selling Price
Classic Cars	\$3,919,616	33,992	\$87.34
Vintage Cars	\$1,903,151	21,069	\$78.15
Motorcycles	\$1,166,388	11,663	\$83.00
Trucks and Buses	\$1,127,790	10,777	\$87.53
Planes	\$975,004	10,727	\$81.74
Ships	\$714,437	8,127	\$83.86
Trains	\$226,243	2,712	\$75.65

2. Deal Size Concentration

Medium-sized deals account for the majority of revenue at **\$6.08M (60.7%)**, with Large deals contributing **\$2.64M (26.3%)** and Small deals just **\$1.30M (13.0%)**. This indicates the business is heavily reliant on its mid-market segment rather than either high-value enterprise deals or high-volume small transactions, worth considering for targeted growth strategies in the Large deal segment.

3. Geographic Distribution

Sales are concentrated in a small number of markets. France leads international performance at \$1.11M, well ahead of other European countries, followed by strong contributions from the USA, UK, and Spain. Many smaller markets (e.g. Ireland, Philippines) contribute under \$100K each, indicating long-tail geographic exposure rather than balanced global distribution.

4. Trend & Time Intelligence

Sales show a clear seasonal, spike-driven pattern rather than steady linear growth, consistent with order-driven B2B sales cycles. Year-over-year comparisons show strong growth into 2004 before a decline toward the end of the covered period. Current period figures were Sales YTD \$896K, Sales QTD \$163K, and Sales MTD \$44K, with a YoY growth measure of 164.8%, reflecting the small transaction base captured in the most recent period of the dataset.

Insights & Recommendations

- **Protect and grow Classic Cars:** as the single largest revenue driver (39% of sales), prioritise inventory availability and marketing spend here first; any disruption to this line has outsized revenue impact.
- **Investigate Trains' underperformance:** determine whether this is a demand issue, a catalogue/SKU gap, or a pricing issue, given it lags every other product line by a wide margin.
- **Develop the Large deal segment:** with Medium deals dominating revenue, a targeted push (account-based selling, bundling) into Large deals could lift average order value meaningfully.
- **Diversify geographic concentration:** reliance on a handful of top markets (France, USA, UK, Spain) represents concentration risk; assess whether under-performing markets are a growth opportunity or structurally limited.
- **Close the profitability gap:** acquiring a cost/COGS data source would allow this analysis to move from revenue-based insights to true margin-based decision-making, which is the natural next step for this dashboard.

Appendix: Dashboard Structure

The accompanying Power BI dashboard (SalesDashboard.pbip) contains four pages:

- **Overview** — headline KPIs, sales trend over time, deal size breakdown, product line and country summaries.
- **Product & Deal Size** — product line performance chart, deal size share, and a detailed product line summary table.
- **Geography** — country-level sales ranking and a detailed country/territory breakdown table.
- **Trends & Time Intelligence** — month-over-month trend, year-over-year comparison, and YTD/QTD/MTD summary cards.

This report and its accompanying dashboard were produced as a portfolio demonstration of AI-assisted BI delivery, built end-to-end with Claude Code and Power BI.